



विद्यया धनं सर्वधनं प्रधानम्

IIT JAMMU

Designing Low Noise Amplifier (LNA) using Open-Source tools



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

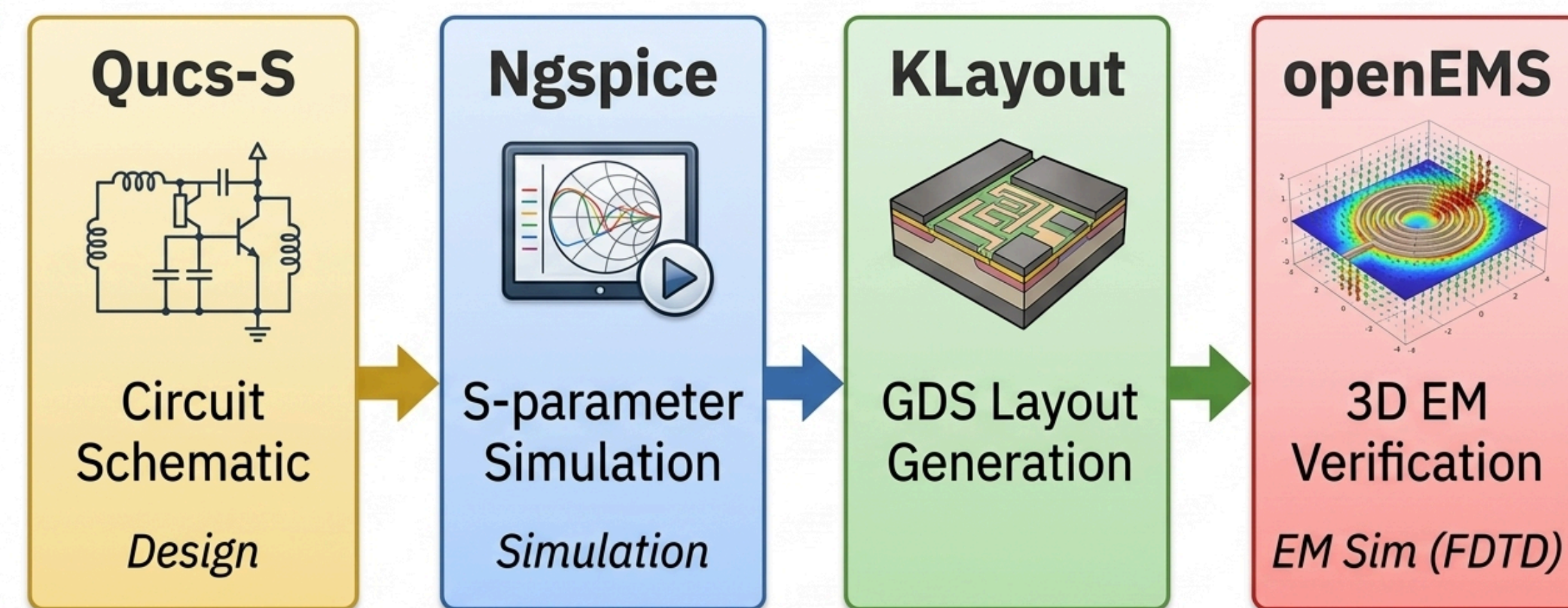
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Problem Statement: The design of mm-wave LNAs at 130 GHz demands precision transistor models and high-frequency passive components typically only accessible through expensive proprietary EDA ecosystems. This project addresses the question: can a fully open-source toolchain reliably reproduce mm-wave LNA performance benchmarks?

Abstract

Low Noise Amplifiers (LNAs) operating at mm-wave frequencies (77–130 GHz) are critical building blocks in 5G, radar, and imaging systems. Commercial EDA tools for such designs are prohibitively expensive, creating barriers for academic research. This work presents the design, simulation, and verification of a 130 GHz LNA using the IHP SG13G2 open-source BiCMOS PDK — entirely within a free, open-source EDA flow comprising Qucs-S, Ngspice, KLayout, and openEMS. S-parameter results (gain, return loss) and Noise Figure are validated against published reference designs, and a Differential Stacked Spiral Inductor (DSSI) is characterized through 3D electromagnetic simulation.



Circuit Topology

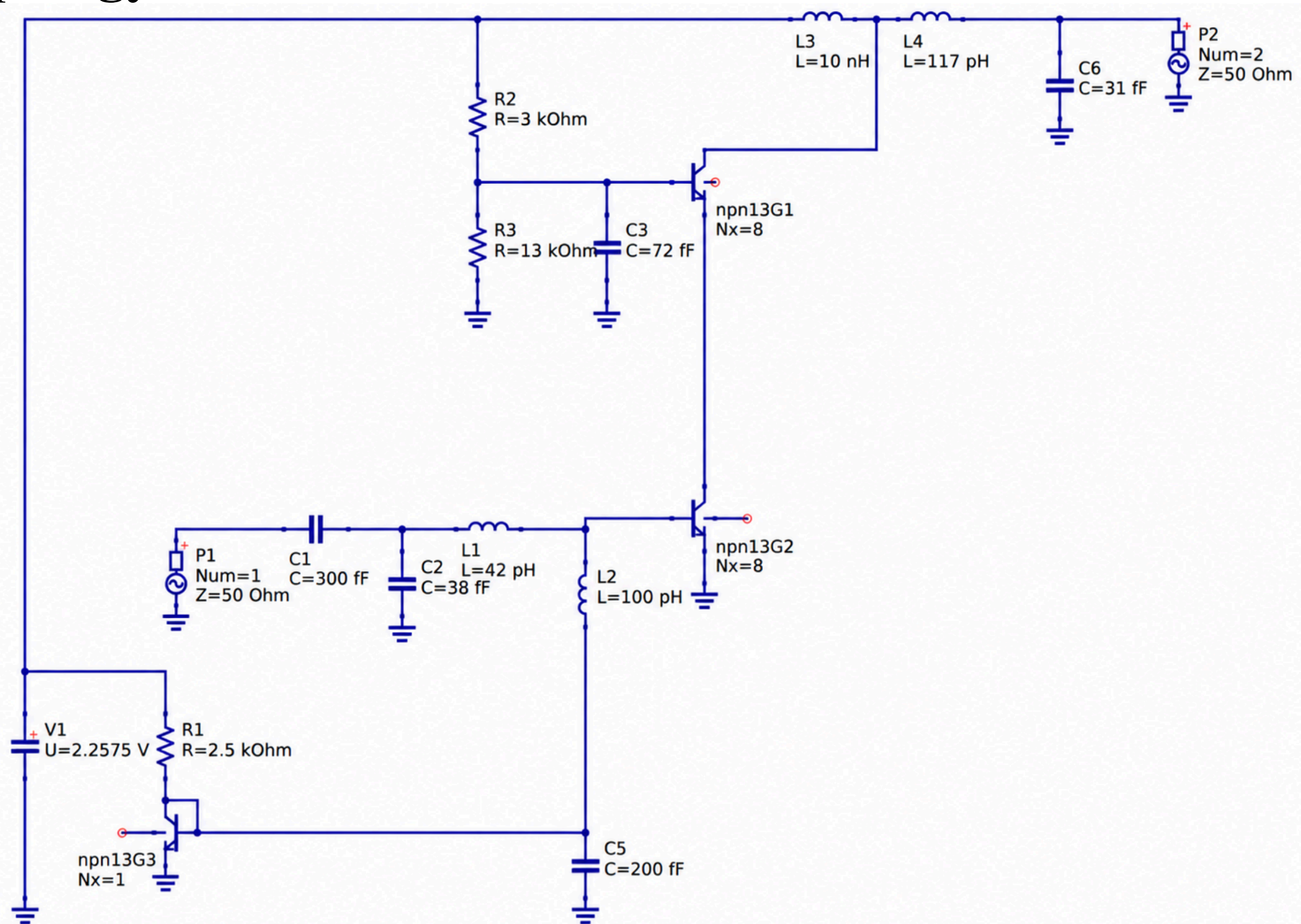


Fig. 1.1 Schematic of the Low Noise Amplifier (LNA) in Qucs-s EDA tool

S - Parameters & Noise Figure

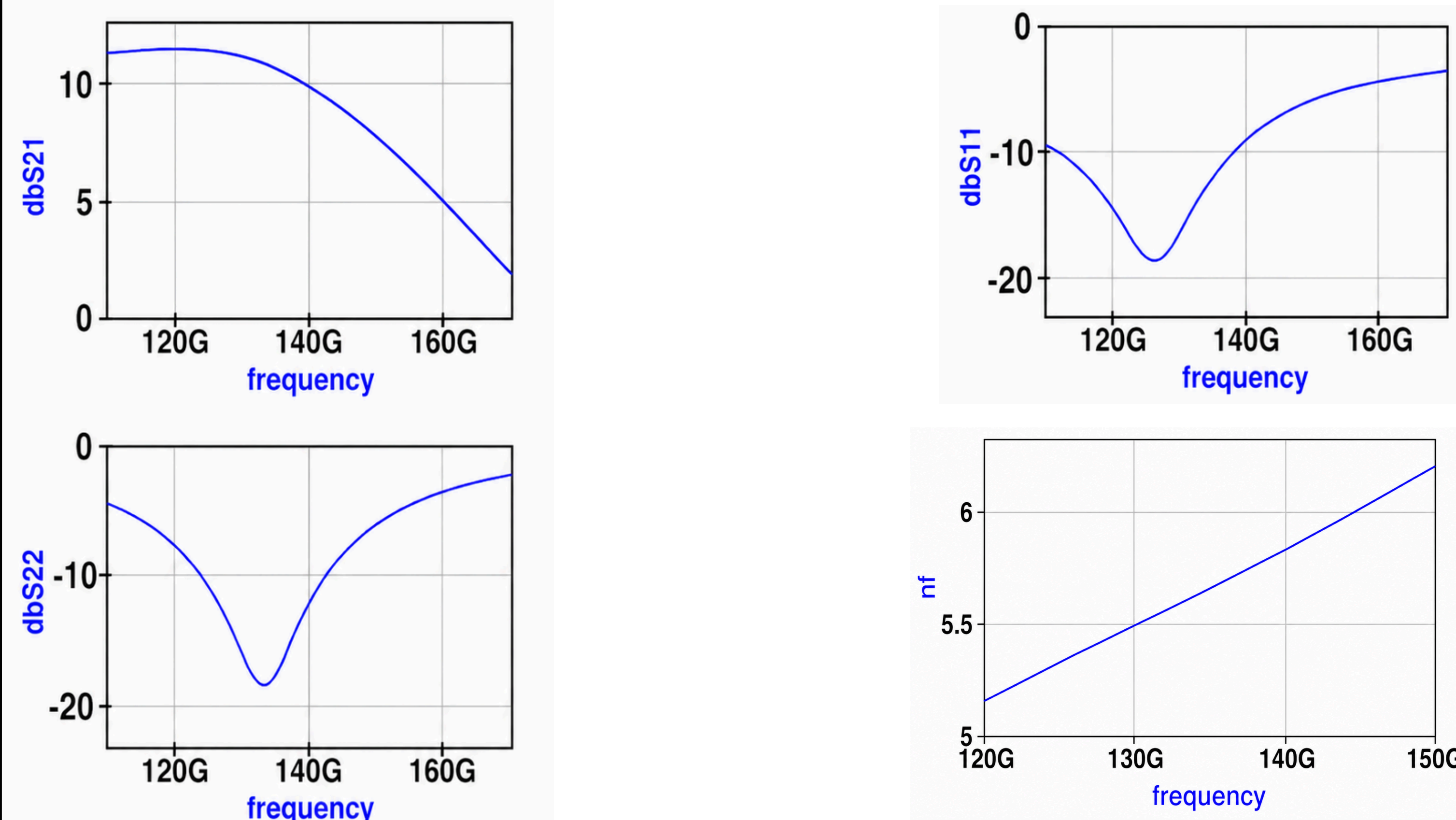


Fig. 1.2 S-parameter & NF Plots in Ngspice

Inductor EM Simulation

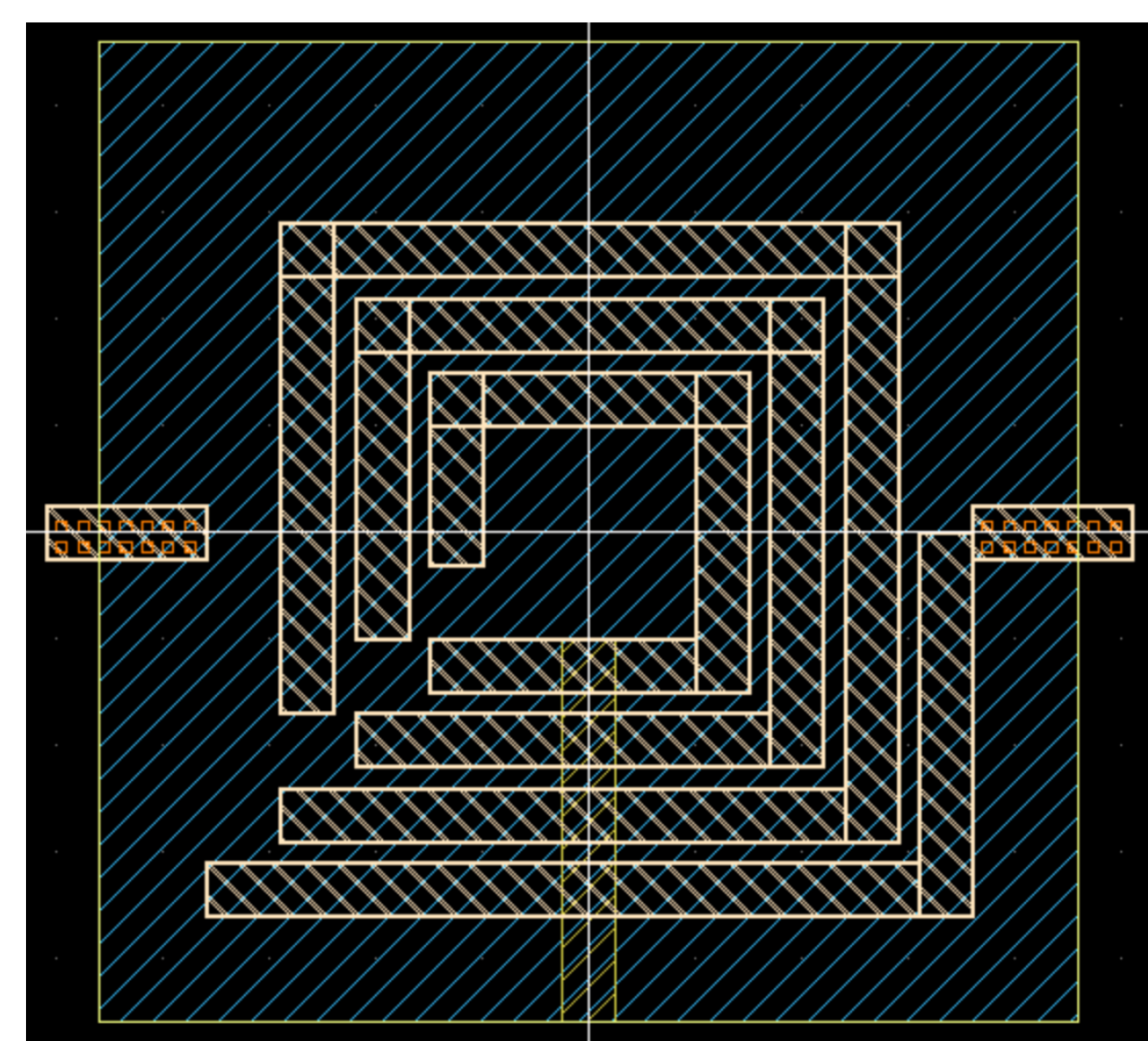


Fig. 1.3 Schematic of the Inductor in KLayout GDS

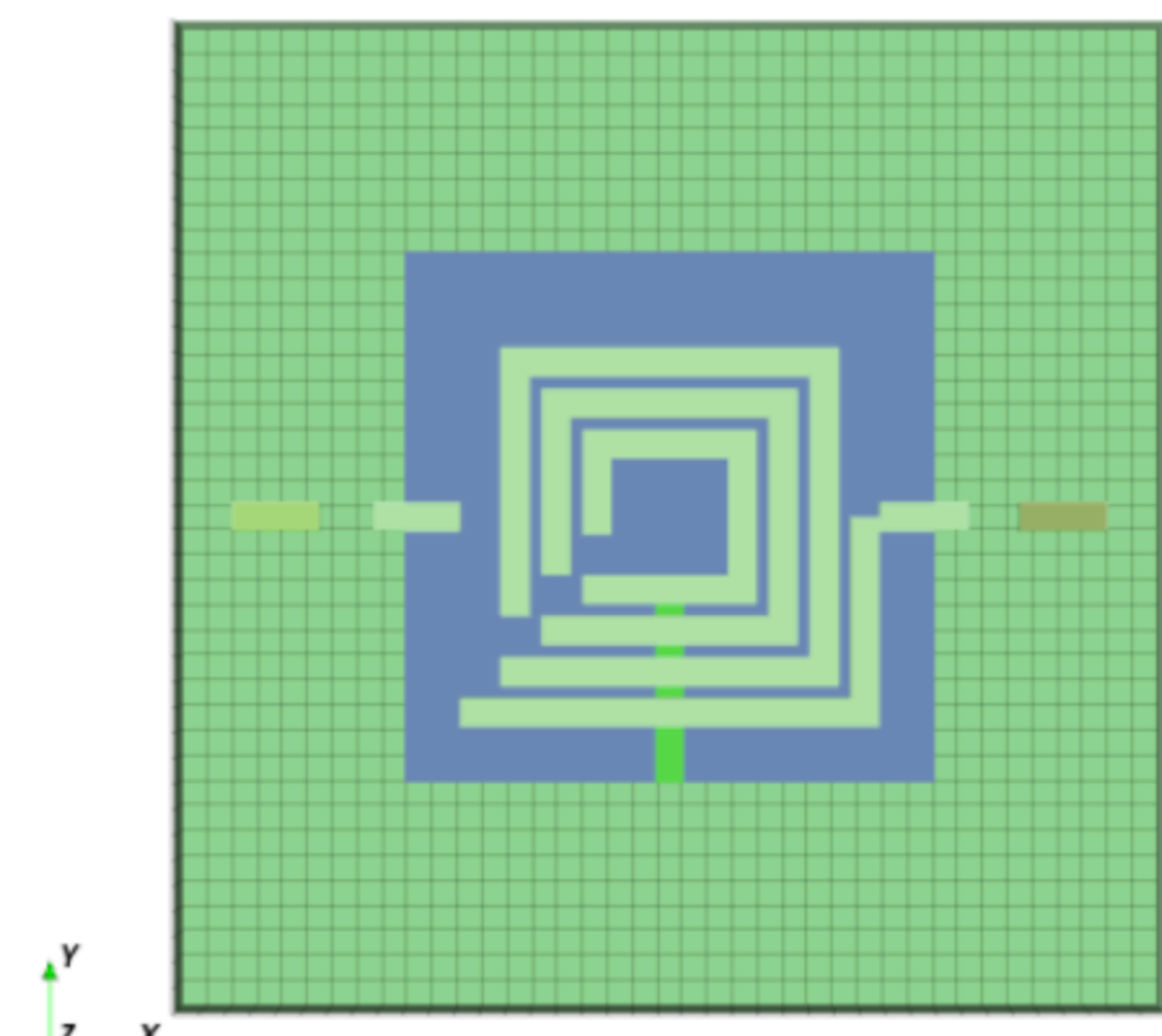


Fig. 1.4 Simulation of the Designed Inductor in OpenEMS

Q-factor / S11 Plot

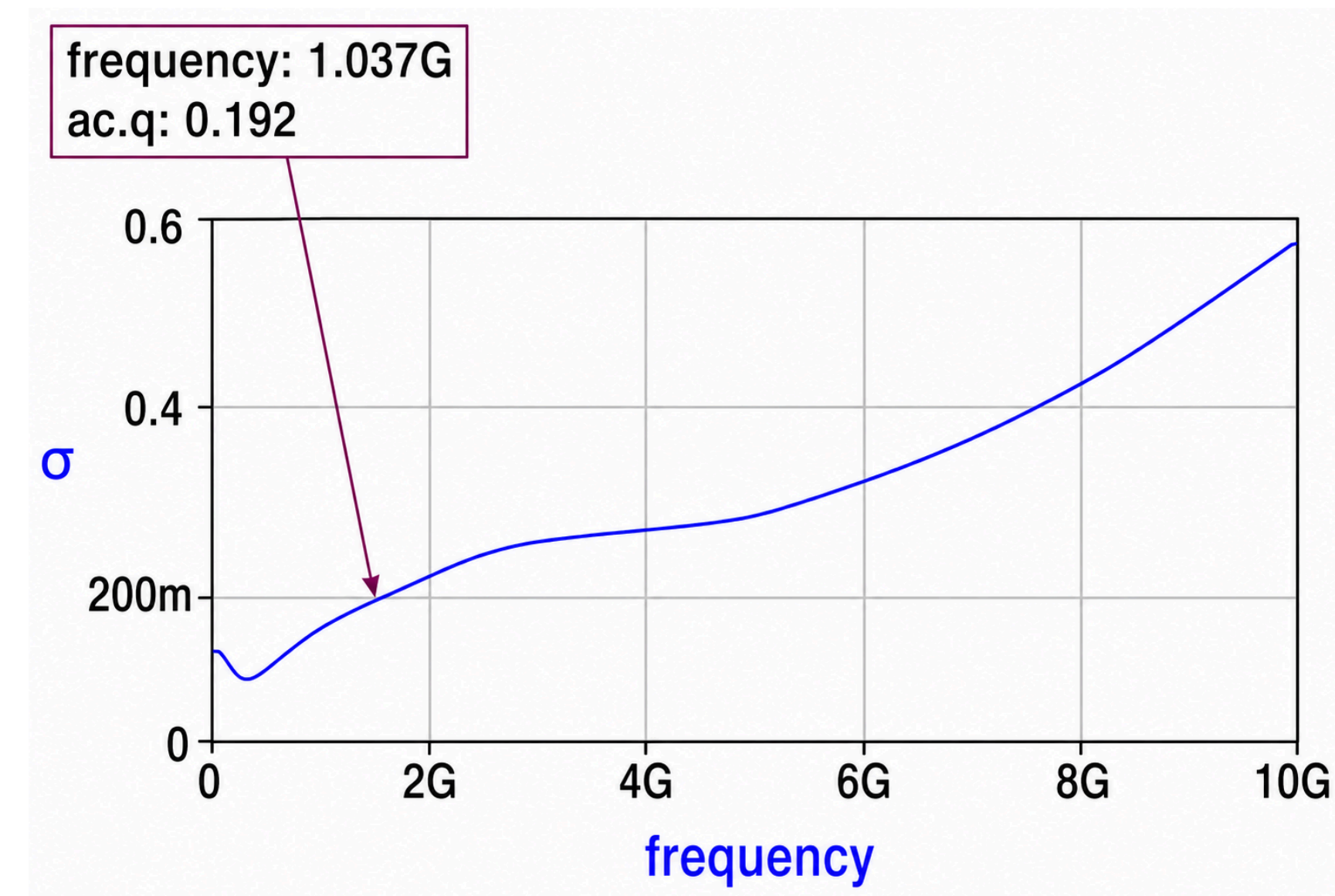


Fig. 1.5 Measure of Proximity to an Ideal Inductor

Key Results Table

Metric	Reference (Paper)	Simulation Studies (Ngspice/Qucs-s)	Agreement (Δ)
S21 (dB)	12.2	12.1	0.1 (Close)
S11 (dB)	-24.1	-20.1	4 (Reasonable)
S22 (dB)	-22.4	-19.8	2.6 (Reasonable)
NF (dB)	5.2	5.5	0.3 (Close)

Conclusion

A complete mm-wave LNA design flow was successfully demonstrated using exclusively open-source tools with the IHP SG13G2 PDK. Simulated S-parameter and Noise Figure results show close agreement with reference literature, validating the open-source approach for 130 GHz circuit design. The work establishes a reproducible, cost-free methodology for mm-wave RFIC research — making advanced analog IC design accessible to academic institutions without commercial tool licenses.

References

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